

Mental Health Data Analysis



Data Wrangling Final Project ## Mental Health & Screen Time Analysis

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Course: Certificate IV in IT (Data Science and AI) — TAFE

Unit: Data Wrangling with Python

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Purpose & Research Questions

This project has two equally weighted objectives:

Objective 1 — Technical: Demonstrate data wrangling competency through the application of Python-based data cleaning, transformation, imputation, functional programming, and visualisation techniques across two structurally different real-world datasets.

Objective 2 — Analytical: Explore mental health through two complementary lenses:

“Among people who self-report mental health symptoms — how severe is depression and anxiety, and who seeks treatment?” (Report 1 — Clinical Survey Data)

“Among people who reach crisis point — who presents to a Western Australian Emergency Department for mental health reasons, and what happens to them?” (Report 2 — WA Emergency Department Data)

Connecting insight: These two datasets cannot be merged — they measure different populations using different instruments in different countries. Their value lies precisely in that difference: Report 1 captures the subclinical self-reported end of the mental health spectrum; Report 2 captures the acute crisis end. The gap between them represents the untreated middle — people who are unwell but reach neither a survey nor an emergency department.

Note on methods: Both reports deliberately explore multiple approaches to the same problem — for example, comparing imputation methods, visualisation types, and analytical framings — to demonstrate that data wrangling decisions are not neutral. Different choices produce different insights.

Datasets

Dataset	Type	Records	Source
Social Media & Mental Health (Braghieri et al., 2022)	Real — validated survey	579	ICPSR — US college students

Dataset	Type	Records	Source
WA Synthetic EDDC	Representative synthetic	~950,000	WA Department of Health, 2022

Datasets Considered but Not Carried Forward

Two Kaggle datasets (*Gaming and Mental Health* and *Social Media Impact on Teen Mental Health*) were initially evaluated but replaced with the WA EDDC dataset, which offers greater analytical depth, local relevance, and institutional credibility as a representative synthetic dataset produced by the WA Department of Health.

Methodology

This project applies the following data wrangling techniques across both reports:

Technique	Report 1	Report 2
Missing value detection & imputation	✓	✓
Data type conversion & encoding	✓	✓
Outlier detection using IQR	✓	✓
Functional programming: <code>map()</code> , <code>filter()</code> , <code>reduce()</code>	✓	✓
Chunked loading for large datasets	✗	✓
Datetime parsing & time-based analysis	✗	✓
PHQ-9 / GAD-7 clinical scale analysis	✓	✗
Visualisation: histograms, boxplots, heatmaps, bar charts	✓	✓

Both reports deliberately explore multiple approaches to the same problem (comparing imputation methods, visualisation types, and analytical framings) to demonstrate that data wrangling decisions are not neutral and that different choices produce different insights.

All analysis is conducted in Python using `pandas`, `matplotlib`, and `seaborn`. Reports are rendered using Quarto and automated via GitHub Actions.

Ethical Considerations

- The Social Media & Mental Health dataset (Braghieri et al., 2022) is openly licensed via ICPSR and contains no personally identifiable information.
- The WA EDDC dataset is a synthetic representative dataset produced by the WA Department of Health. It contains no real patient records: synthetic instances close to source data have been removed to protect patient privacy.
- Neither dataset contains personally identifiable information.

- AI tools (Claude) were used as a thinking partner during project planning and code development. All analysis decisions, interpretations, and written content are the author's own.
- AI tool use is disclosed in accordance with TAFE academic integrity guidelines.